

# Test Plan 7 — External API Monitoring (Govmap + Ofek)

|                |                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project        | monitoring (opt-in — registered only when MONITORING_ENABLED=true )                                                                                                                                                                 |
| Specs          | tests/monitoring/govmap.spec.ts , tests/monitoring/ofek.spec.ts                                                                                                                                                                     |
| Default filter | none — every matched spec runs                                                                                                                                                                                                      |
| Target         | Live third-party map dependencies: Govmap ( www.govmap.gov.il ) and Ofek ( basemaps.govmap.gov.il )                                                                                                                                 |
| Client         | APIRequestContext only (no browser); endpoints from tests/monitoring/support/endpoints.ts ( config.json → monitoring , env-overrideable)                                                                                            |
| Cases          | 50 (Govmap 25 + Ofek 25)                                                                                                                                                                                                            |
| Skips          | Block page → skip: a spec skips when the govmap.gov.il edge serves an HTML block/challenge page to the runner (a geo/bot block). Otherwise none — not in the default suite; runs only when opted in, and a real break IS the alert. |
| Skill          | organuz-monitoring                                                                                                                                                                                                                  |

## Scope

Dedicated availability and contract monitoring for the two external map services the product relies on when it characterizes a property. Both are Survey-of-Israel / national GIS (geographic information system) services:

- Govmap ( www.govmap.gov.il ) — the map API and the address search/geocoding ("geocoding" means turning an address into map coordinates).
- Ofek ( basemaps.govmap.gov.il ) — the national orthophoto "אופק" aerial tiles that the roof scan runs on, plus the label/line overlay tiles. ("Orthophoto" tiles are aerial photos corrected so they can be used like a map.)

Both services have caused product incidents when they broke, so this group acts as the early-warning signal.

These tests are **health checks**: when a dependency breaks, they fail, and that failure is itself the alert. Because they hit live third-party APIs — and would go red on someone else's outage — the

group is **opt-in** ( `MONITORING_ENABLED=true` ) and runs on a schedule. It is **never** part of the PR green gate. The default `npx playwright test` never registers the `monitoring` project, so a Govmap or Ofek outage can't break the PR suite.

## Gating (sanctioned skip — block page)

The one sanctioned skip is a **block/challenge page**. The `govmap.gov.il` edge sometimes serves an HTML block or challenge page to the runner — a geo/bot block, for example when a CI runner sits outside Israel. That is not a Govmap outage and must not read as one, so a blocked run **skips** instead of failing.

- A per-worker canary in `tests/monitoring/support/availability.ts` ( `govmapBlockReason` / `ofekBlockReason` ) runs from each spec's `beforeEach` . When it detects an HTML block/challenge page, the spec calls `test.skip` with a clear reason.
- A **real break still FAILS**: a 5xx, a connection error, or a real-but-wrong payload (a JSON/tile response with the wrong shape — *not* an HTML page) all go red. So a red monitoring run now means a genuine Govmap/Ofek break, not a blocked runner.

## Preconditions

- `MONITORING_ENABLED=true` in the environment. Otherwise the project is not registered at all (see `playwright.config.ts` ).
- Public internet access to `www.govmap.gov.il` and `basemaps.govmap.gov.il` .
- No credentials, no browser, and no password gate — the specs use `APIRequestContext` (Playwright's plain HTTP client) directly. The endpoint hosts, the public embed token, the known address, the tile layer names, the sample tile coordinate, and the latency budget all come from `config.json → monitoring` , and each can be overridden by an environment variable.

## Cases

### `govmap.spec.ts` — "Govmap API monitoring" ( `@monitoring` )

Epic: External API monitoring · Feature: Govmap ( `www.govmap.gov.il` )

| ID     | CASE                                           | ASSERTS                   |
|--------|------------------------------------------------|---------------------------|
| GOV-01 | <code>govmap.api.js</code> loader is reachable | Loader script returns 200 |

| ID     | CASE                                                 | ASSERTS                                                                        |
|--------|------------------------------------------------------|--------------------------------------------------------------------------------|
| GOV-02 | <code>govmap.api.js</code> is served as JavaScript   | <code>content-type</code> matches <code>javascript</code>                      |
| GOV-03 | <code>govmap.api.js</code> is a non-trivial bundle   | Body > 100 KB and contains <code>govmap</code>                                 |
| GOV-04 | Loader responds within the latency budget            | <code>res.ok()</code> and elapsed < <code>LATENCY_BUDGET_MS</code>             |
| GOV-05 | Govmap is served over HTTPS                          | API base URL starts with <code>https://</code>                                 |
| GOV-06 | The embed viewer document loads for the public token | Embed URL returns 200                                                          |
| GOV-07 | The embed viewer document is HTML                    | <code>content-type</code> matches <code>text/html</code>                       |
| GOV-08 | Address autocomplete responds 200                    | POST autocomplete returns 200                                                  |
| GOV-09 | Address autocomplete returns JSON                    | <code>content-type</code> matches <code>application/json</code>                |
| GOV-10 | A known address returns at least one result          | <code>resultsCount &gt; 0</code> and <code>results</code> is a non-empty array |
| GOV-11 | Top result matches the queried street                | <code>results[0].text</code> contains the known street                         |
| GOV-12 | Top result carries a POINT geometry (geocoded)       | <code>results[0].shape</code> matches <code>POINT(...)</code>                  |
| GOV-13 | Top result is typed as an address                    | <code>results[0].type === 'address'</code>                                     |
| GOV-14 | Autocomplete respects <code>maxResults</code>        | <code>results.length &lt;= 10</code>                                           |
| GOV-15 | Autocomplete responds within the latency budget      | <code>res.ok()</code> and elapsed < <code>LATENCY_BUDGET_MS</code>             |
| GOV-16 | A nonsense query is handled gracefully               | No 5xx; <code>results</code> is an array; <code>resultsCount === 0</code>      |

| ID     | CASE                                                       | ASSERTS                                                           |
|--------|------------------------------------------------------------|-------------------------------------------------------------------|
| GOV-17 | Address search is HTTPS-only                               | Autocomplete URL starts with <code>https://</code>                |
| GOV-18 | <code>getTypes</code> responds 200 with a JSON array       | 200 and body is an array                                          |
| GOV-19 | <code>getTypes</code> exposes the known search types       | Type names include <code>settlement</code> and <code>layer</code> |
| GOV-20 | The embed token is still authorized for the app host       | Auth handshake (token + host URL) returns 200                     |
| GOV-21 | A protected layers-catalog endpoint enforces auth          | <code>baseLayers</code> without a session returns 400/401/403     |
| GOV-22 | Anonymous <code>users-management/me</code> is unauthorized | Returns 401                                                       |
| GOV-23 | The Hebrew translations bundle is available                | Returns 200 with a JSON <code>content-type</code>                 |
| GOV-24 | No critical Govmap endpoint returns a 5xx                  | Loader, embed, autocomplete, <code>getTypes</code> all < 500      |
| GOV-25 | The Govmap host is reachable over TLS                      | A valid-cert HTTPS request to the base URL returns < 500          |

## `ofek.spec.ts` — "Ofek orthophoto tiles monitoring" ( `@monitoring` )

Epic: External API monitoring · Feature: Ofek / Survey-of-Israel orthophoto

( `basemaps.govmap.gov.il` )

The Ofek tile server only serves requests that come from the Govmap origin, so every tile request carries that origin in its `Referer` header (set via `TILE_HEADERS` ).

| ID     | CASE                                                      | ASSERTS                                                   |
|--------|-----------------------------------------------------------|-----------------------------------------------------------|
| OFK-01 | An orthophoto tile responds 200 (with the Govmap referer) | Ortho tile returns 200                                    |
| OFK-02 | The orthophoto tile is served as <code>image/jpeg</code>  | <code>content-type</code> matches <code>image/jpeg</code> |
| OFK-03 | The orthophoto tile body is non-empty                     | Body length > 1000                                        |

| ID     | CASE                                                          | ASSERTS                                                                                                 |
|--------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| OFK-04 | The orthophoto tile has valid JPEG magic bytes                | Body starts with <code>FF D8 FF</code>                                                                  |
| OFK-05 | The orthophoto tile responds within the latency budget        | <code>res.ok()</code> and elapsed < <code>LATENCY_BUDGET_MS</code>                                      |
| OFK-06 | Orthophoto tiles are served over HTTPS                        | Tiles base URL starts with <code>https://</code>                                                        |
| OFK-07 | The tile server requires the Govmap referer                   | Request without a referer returns 401                                                                   |
| OFK-08 | The tile server rejects a foreign referer                     | A foreign <code>Referer</code> returns 401                                                              |
| OFK-09 | Adjacent orthophoto tiles are available                       | <code>(x+1,y)</code> and <code>(x,y+1)</code> both return 200                                           |
| OFK-10 | A 2x2 orthophoto tile grid is fully available                 | All four tiles return 200                                                                               |
| OFK-11 | A deeper-zoom orthophoto tile (roof-scan detail) is available | The <code>z+2</code> child tile returns 200                                                             |
| OFK-12 | The orthophoto tile size is within a sane range               | Body between 1 KB and 2 MB (not an error page)                                                          |
| OFK-13 | The orthophoto tile advertises CDN caching                    | One of <code>cache-control</code> / <code>etag</code> / <code>age</code> / <code>expires</code> present |
| OFK-14 | A label overlay tile responds 200 (with the referer)          | Label tile returns 200                                                                                  |
| OFK-15 | The label tile is served as <code>image/png</code>            | <code>content-type</code> matches <code>image/png</code>                                                |
| OFK-16 | The label tile has valid PNG magic bytes                      | Body starts with <code>89 50 4E 47</code>                                                               |
| OFK-17 | The label tile body is non-empty                              | Body length > 200                                                                                       |
| OFK-18 | The label tile responds within the latency budget             | <code>res.ok()</code> and elapsed < <code>LATENCY_BUDGET_MS</code>                                      |

| ID     | CASE                                                             | ASSERTS                                                                 |
|--------|------------------------------------------------------------------|-------------------------------------------------------------------------|
| OFK-19 | The label tile also requires the Govmap referer                  | Request without a referer returns 401                                   |
| OFK-20 | Ortho and label tiles are both available for the same coordinate | Both return 200                                                         |
| OFK-21 | The Ofek tiles host is reachable over TLS                        | A valid-cert HTTPS tile request returns < 500                           |
| OFK-22 | Orthophoto tiles return no server errors across a sample         | Four sampled zoom/coords all < 500                                      |
| OFK-23 | The configured orthophoto layer still serves imagery             | Ortho tile 200 — guards a silent layer rename                           |
| OFK-24 | The configured labels layer still serves tiles                   | Label tile 200 — guards a silent layer rename                           |
| OFK-25 | A batch of orthophoto tiles are all <code>image/jpeg</code>      | All four tiles' <code>content-type</code> match <code>image/jpeg</code> |

## Run

```
# Opt in explicitly (the project is unregistered otherwise):
npm run test:monitoring
# equivalently:
MONITORING_ENABLED=true npx playwright test --project=monitoring
```

## Scheduling & alerting

This group runs on its own GitHub Actions workflow, `External API Monitoring` ( `.github/workflows/monitoring.yml` ). It is kept separate from the PR suite ( `parallel-tests.yml` ) so that an outage alerts here without ever failing the green PR gate:

- **Schedule:** cron `*/30 * * * *` (every 30 minutes). The `concurrency` setting cancels an in-flight run so a newer pass supersedes it.
- **Runner:** `ubuntu-latest`, Node 22, `npm ci`, and no browser install (the specs use `APIRequestContext` only). It runs `npx playwright test --project=monitoring` with `MONITORING_ENABLED=true`.

- **Alert issue:** on failure, the workflow opens a single auto-managed GitHub issue (label `monitoring-alert`) — or, if one is already open, comments "still failing" on it. That way a sustained outage doesn't spam a new issue on every run. The next green run comments on the issue and closes it (auto-recovery).
- **Slack:** on failure, it posts to every configured webhook — `SLACK_WEBHOOK_URL` and `SLACK_WEBHOOK_BOT_URL` (GitHub repo secrets). Each is independent and optional: an unset webhook is skipped, and a failed post is non-fatal, so one bad webhook never masks the alert.
- **Alert-path test:** a `workflow_dispatch` run with `force_fail=true` fails on purpose. This exercises the issue + Slack path without hitting the live APIs; the next green run auto-closes the resulting issue.

## Notes

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- This group is **never** in the default suite. It either runs (when opted in) or is unregistered. Its one sanctioned skip is a **block/challenge page** from the govmap.gov.il edge (a geo/bot block — see "Gating" above): the per-worker canary in `tests/monitoring/support/availability.ts` skips the spec via each `beforeEach`. Every other red is a real signal that a critical map dependency is degraded — a 5xx, a connection error, or a real-but-wrong (non-HTML) payload is a break, not a flake to skip.
- The endpoint hosts, the public embed token, the tile layer names, the sample tile, the known address, and the latency budget all live in `config.json → monitoring`. Each can be overridden by an environment variable ( `GOVMAP_API_URL`, `OFEK_TILES_URL`, `OFEK_ORTHO_LAYER`, `OFEK_LABELS_LAYER`, `MONITORING_LATENCY_MS`, and so on). The layer-drift guards (OFK-23/24) exist because the national orthophoto layer is renamed every year; a rename is a real signal, and it is fixed with a one-line config update.
- Adding or removing a monitoring spec changes the documented count (50); update `CLAUDE.md`, this plan, and the `test-suite-parity` skill.